

Ultrafast Optical Nanoscopy of Carrier Dynamics in Silicon Nanowires

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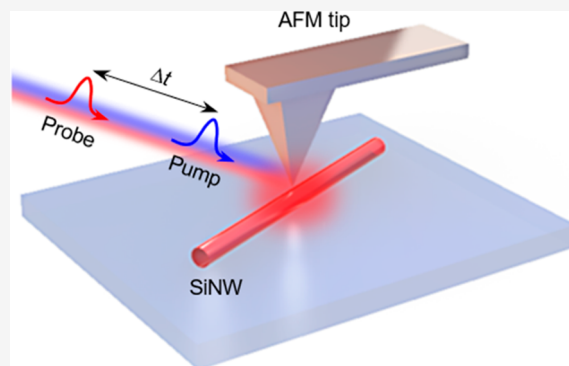
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Supporting Information

ABSTRACT: Carrier distribution and dynamics in semiconductor materials often govern their physical properties that are critical to functionalities and performance in industrial applications. The continued miniaturization of electronic and photonic devices calls for tools to probe carrier behavior in semiconductors simultaneously at the picosecond time and nanometer length scales. Here, we report pump–probe optical nanoscopy in the visible-near-infrared spectral region to characterize the carrier dynamics in silicon nanostructures. By coupling experiments with the point-dipole model, we resolve the size-dependent photoexcited carrier lifetime in individual silicon nanowires. We further demonstrate local carrier decay time mapping in silicon nanostructures with a sub-50 nm spatial resolution. Our study enables the nanoimaging of ultrafast carrier kinetics, which will find promising applications in the future design of a broad range of electronic, photonic, and optoelectronic devices.

KEYWORDS: silicon nanowires, s-SNOM, pump probe, carrier dynamics, nanoscopy



Semiconductor technology has stimulated the exponential progress of microelectronics in the last half-century. Silicon-based complementary metal-oxide-semiconductor integrated circuits form the backbone of modern technology.^{1,2} In addition to the prevalence in electronic devices, silicon nanostructures are widely exploited in photonics for versatile control of light propagation, detection, and modulation.^{3–5} Recently, emerging low-dimensional materials, particularly silicon nanowires (SiNWs), have also opened new possibilities for broad applications in electronics, photonics, photovoltaics, and photoelectrochemistry.^{6–9}

For most semiconductor materials, free-carrier concentration determines their electrical and optical properties, such as conductivity, refractive index, and absorption coefficients.^{10–12} In addition, the performance of transistors and semiconductor devices primarily depends on the spatial and temporal carrier distribution.^{13–16} Therefore, resolving the charge transport and carrier dynamics in semiconductor nanostructures is essential for designing future electronic and photonic devices.

Time-resolved terahertz spectroscopy has been widely applied to reveal carrier dynamics in bulk and nanostructured semiconductors.^{17,18} In addition, pump–probe microscopy was developed to probe the site-specific carrier transport and recombination.^{19–21} However, the spatial resolution of this microscopic method is limited by optical diffraction. Alternatively, tip-based scattering-type scanning near-field optical microscopy (s-SNOM) is capable of mapping free-

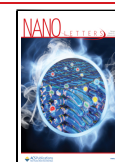
carrier distribution in semiconductor nanostructures with a nanoscale resolution.^{22,23} The spatial resolution is defined by the radius of the tip apex. Ultrafast infrared-terahertz nanospectroscopy was further developed through the integration of s-SNOM and pump–probe optics. However, given the limited photon energy, the efforts have been primarily focused on studying carrier dynamics in narrow bandgap semiconductors, such as InAs^{24,25} and Hg_{1–x}Cd_xTe,²⁶ or graphene plasmon.²⁷

Here, we report near-field ultrafast optical nanoscopy in the visible-near-infrared spectral region to access the carrier dynamics in silicon, one of the most prevalent materials in current semiconductor technology. Our pump beam has a wavelength of 400 nm (3.1 eV), which is sufficient to excite carriers in common optoelectronic semiconductors, including silicon (bandgap of 1.12 eV) and GaAs (bandgap of 1.42 eV). By combining ultrafast nanoscale measurements and theoretical modeling, we unravel the local photocarrier recombination dynamics in silicon nanowires. Moreover, we demonstrate the spatial mapping of carrier lifetime in silicon with a sub-50 nm

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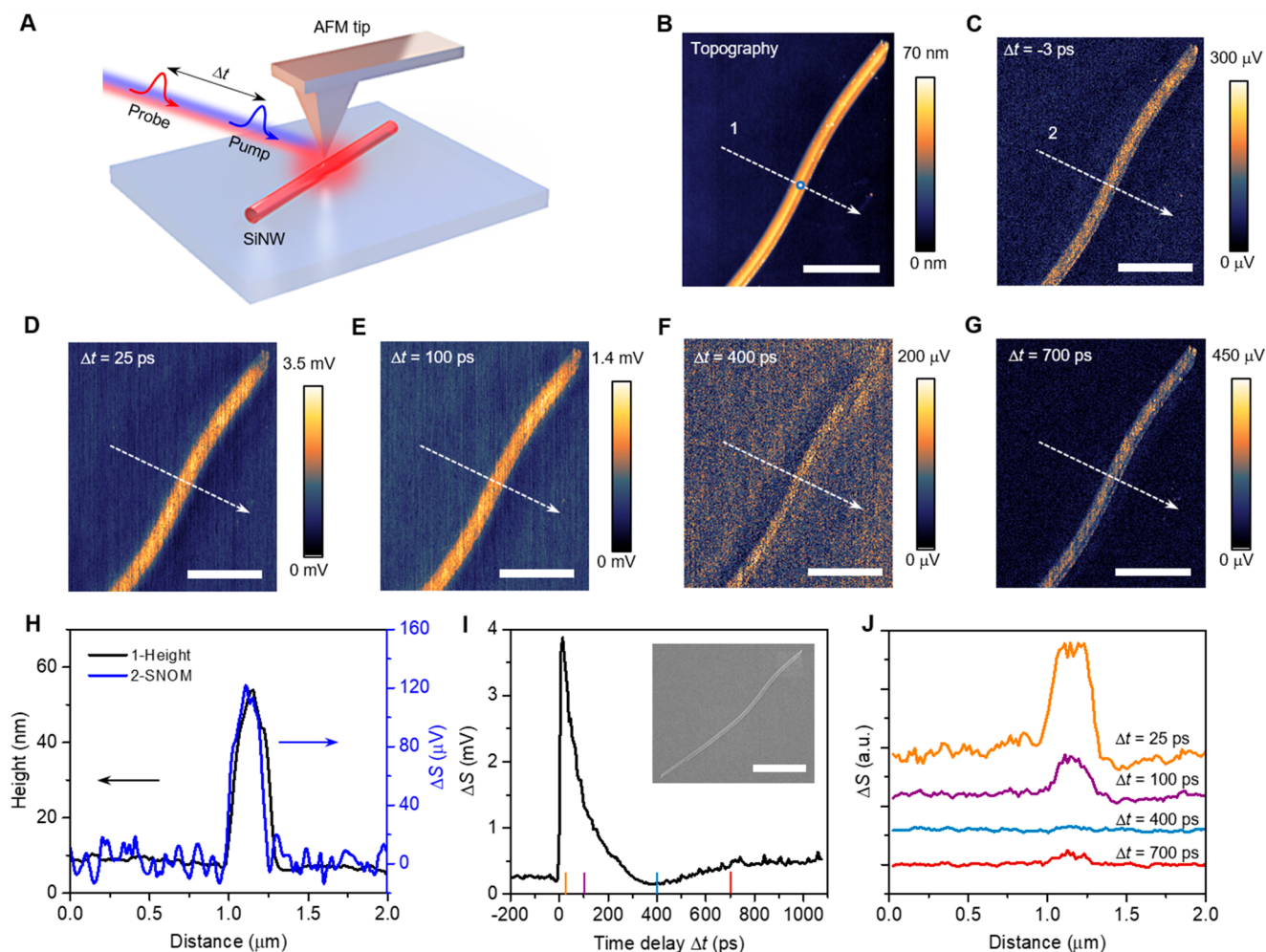


Figure 1. Time-resolved near-field imaging of a SiNW. (A) Schematic of the pump–probe optical nanoscopy setup. The wavelengths of the pump and probe beams are 400 and 800 nm, respectively. (B) AFM topography of a SiNW. (C to G) Time-resolved s-SNOM imaging of the SiNW with a pump–probe delay time of (C) -3 ps, (D) 25 ps, (E) 100 ps, (F) 400 ps, and (G) 700 ps. (H) Height and s-SNOM profiles along the dashed lines in (B) and (C). (I) Transient s-SNOM signal as a function of delay time measured at the SiNW (marked by the dot in (B)). Inset shows the SEM image of the SiNW. (J) s-SNOM line profiles across the SiNW at different pump–probe delay times in (D) to (G). Scale bars: (B to G) 1 μm , Inset in (I): 2 μm .

resolution. Our results provide the capability to probe carrier behaviors in nanoscale materials and devices, which is of great significance to understanding the optoelectronic properties and practical functionality of semiconductor nanostructures.

Figure 1A shows the schematic of our pump–probe near-field optical nanoscopy. A 400 nm pump beam and an 800 nm probe beam with a controlled delay time Δt are directed to an atomic force microscope (AFM) tip with an oscillating frequency Ω (see Figure S1 for more details on the setup). The backscattered probe beam is detected and demodulated at the third harmonic of the tapping frequency to suppress the background noise.^{28,29} The samples in this study are chemically synthesized SiNWs (see inset in Figure 1I). Figure 1B and Figure 1C–G show the topography of a SiNW and the time-resolved s-SNOM imaging at different time delays, respectively. The high spatial overlap of the height and s-SNOM profiles indicates the successful detection of near-field signals (Figure 1H). The time-dependent snapshots distinctly exhibit a dynamic change in the intensity of scattered light (Figure 1J). Specifically, the s-SNOM signal amplitude undergoes an intense increase (Figure 1D), followed by a decay process (Figure 1E and F) and a slow rise at longer delay

times (Figure 1G). This result can also be quantitatively visualized in the transient curve presented in Figure 1I.

To understand the evolution of the near-field scattering signal over the increasing delay time, we adopt a point-dipole model to analyze the scattered light (see Note S1).^{23,30,31} Briefly, the scattered field E_{sc} by the tip is derived from the dipolar near-field interaction between the AFM tip and the sample with the dielectric function ϵ , yielding^{30,31}

$$\mathbf{E}_{\text{sc}} \propto A e^{i\varphi} \mathbf{E}_{\text{in}} = \frac{\alpha(1 + \beta)}{1 - \frac{\alpha\beta}{16\pi(a+z)^3}} \mathbf{E}_{\text{in}} \quad (1)$$

Here the measured scattering intensity $S \propto A^2$, φ is the phase of scattered light, a is the AFM tip radius, z is the tip–sample distance, $\alpha = 4\pi a^3$ is the polarizability of the tip, and $\beta = (\epsilon - 1)/(\epsilon + 1)$, respectively (Figure 2A). The dielectric function ϵ is further calculated as a function of free carrier density N based on the Drude model (Figure S2)³²

$$\epsilon(\omega, N) = \epsilon_{\infty} - \frac{\omega_p^2}{\omega(\omega + i\Gamma)} \quad (2)$$

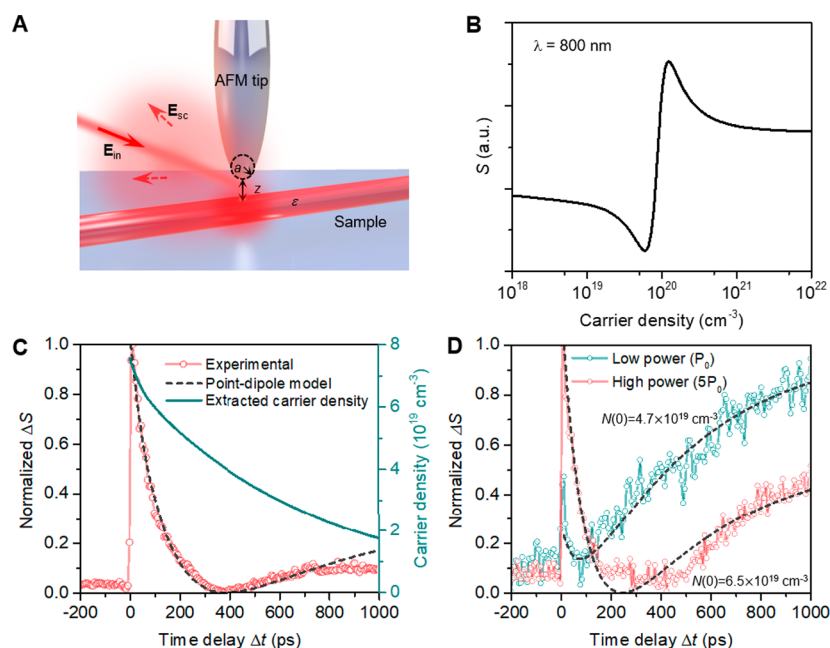


Figure 2. Point-dipole modeling. (A) Schematic of the point-dipole model. a , z , and ϵ are the AFM tip radius, tip–sample distance, and the permittivity of the sample, respectively. (B) Modeled s-SNOM intensity as a function of carrier density in SiNWs. (C) Experimental transient s-SNOM signal, fitting curve with the point-dipole model, and the extracted carrier density in SiNW as a function of pump–probe delay time. (D) Experimental transient s-SNOM scans and fitting curves under different pump excitation powers.

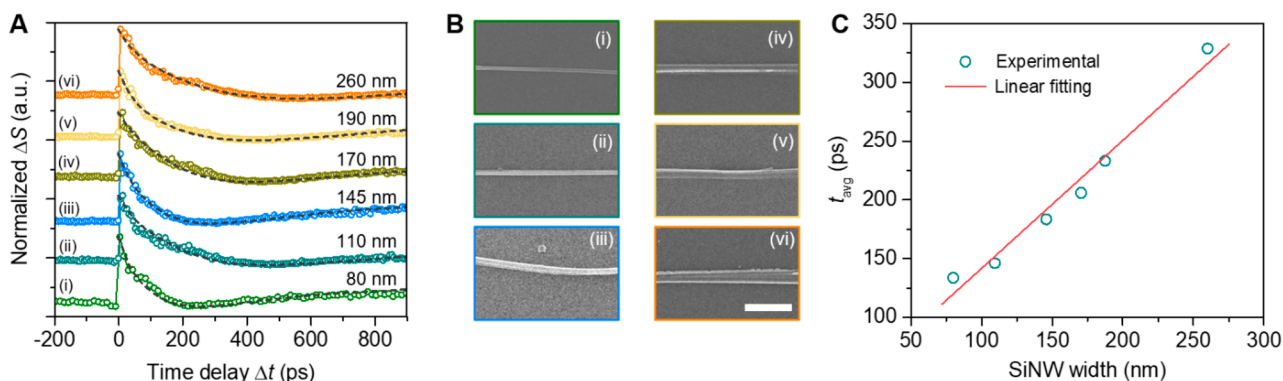


Figure 3. Size-dependent carrier dynamics. (A) Experimental transient s-SNOM scans measured from SiNWs with different widths (i) to (vi). The gray dashed curves present the fitting results by the point-dipole model. (B) Corresponding SEM images of the SiNWs in (A). (C) Size-dependent average decay time and the linear fitting. Scale bar: (B) 1 μm .

where $\epsilon_\infty = 11.7$ for silicon, $\omega_p(N)$ is the plasma frequency and a function of the carrier concentration N (see Note S2), and Γ is the damping rate. In addition, the effect of temperature is considered by the Jellison-Modine model (see Figure S3 and Note S2 for more details).³³

Figure 2B shows the calculated s-SNOM intensity with an increasing free-carrier concentration N from 10^{18} to 10^{22} cm^{-3} at the wavelength of 800 nm. As N increases, the calculated s-SNOM signal exhibits a minimum at $N \sim 6 \times 10^{19} \text{ cm}^{-3}$ and then reaches its maximum at $N \sim 1.2 \times 10^{20} \text{ cm}^{-3}$. This behavior arises from the resonant near-field interaction between the AFM probing tip and the free carriers in SiNWs.³⁰ Different from far-field transient absorption measurements, when the carrier concentration decreases to below $N \sim 6 \times 10^{19} \text{ cm}^{-3}$, the s-SNOM signal increases. This result is consistent with the signal rise at the longer delay time observed in our experimental data. The transient s-SNOM signal can be well fitted by the point-dipole model assuming a biexponential

decay of photoexcited carriers, i.e., $N(\Delta t) = N(0) \times (A_1 e^{-\Delta t/t_1} + A_2 e^{-\Delta t/t_2})$ (Figure 2C). The biexponential kinetics (t_1 , t_2) are attributed to the carrier recombination and diffusion,³⁴ and the average carrier lifetime t_{avg} can be obtained by $t_{\text{avg}} = (A_1 + A_2)/(A_1/t_1 + A_2/t_2)$. We exclude the effect of laser heating as the temperature increase is negligible with respect to the s-SNOM amplitude (Note S3 and Figure S4). In addition, the variation of tip–sample distance due to thermal expansion of SiNWs only leads to slight changes in the scattered light intensity (Note S4 and Figure S5). These results verify the dominant role of carrier density in the transient s-SNOM measurements. We further validate the point-dipole model by measuring the same SiNW with different pump power (Figure 2D), where two curves can be well fitted with different initial carrier densities and the same decay kinetics (Table S1).

We then apply pump–probe s-SNOM to investigate the carrier dynamics in individual SiNWs with varying geometries. Figure 3A shows the transient s-SNOM signals measured in

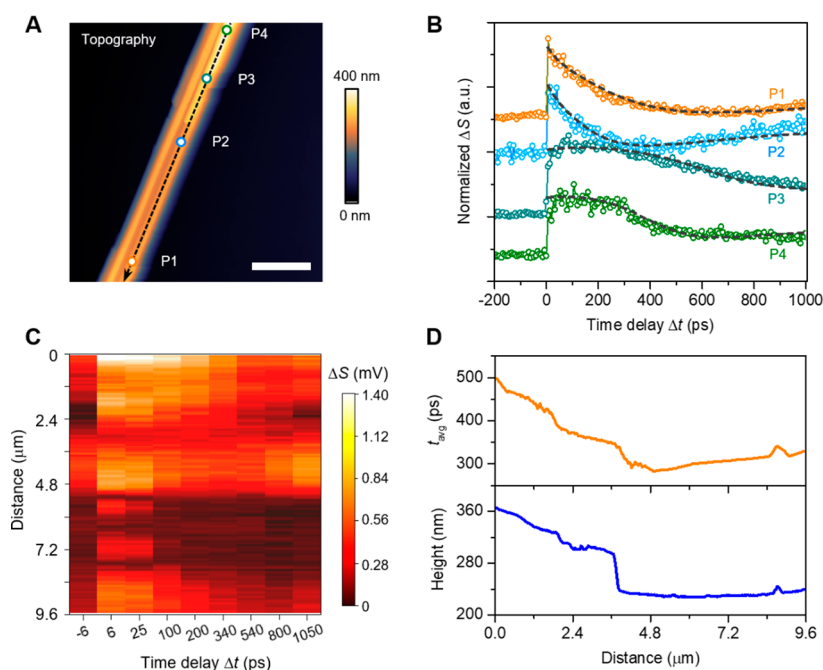


Figure 4. Probing spatially resolved carrier dynamics at the nanoscale. (A) AFM topography of a nonuniform silicon nanowire structure. (B) Experimental transient s-SNOM scans and fitting curves measured at different locations, as marked in (A). (C) Time-resolved s-SNOM mapping along the dashed line in (A). (D) Corresponding decay time and height profiles with a spatial resolution of 35 nm. Scale bar: (A) 2 μm .

different nanowires with various widths. The corresponding scanning electron microscope (SEM) images are shown in Figure 3B. All experimental curves are well fitted with the point-dipole model, and the decay times are summarized in Figure 3C. The carrier lifetime shows a linear increase with the increasing size of the SiNWs, indicating that surface recombination dominates in semiconductor nanowires.³⁵ The surface recombination velocity (SRV) can be calculated from carrier lifetime t_{avg} as $\text{SRV} = d/4t_{\text{avg}}$, where d is the SiNW width.³⁶ The linear fitting gives a surface recombination velocity of 2.2×10^4 cm/s, which is consistent with previous reports.^{34,37} These results validated that our tool and model are effective to probe the carrier lifetime in semiconductor nanomaterials.

We further explore the capability of near-field ultrafast nanoscopy to probe the spatially resolved carrier dynamics with a nanoscale resolution, which cannot be accessed with conventional pump–probe microscopy. A nonuniform silicon nanowire is selected as the test sample (Figure 4A). Transient s-SNOM signals measured at different locations (P1–P4 in Figure 4A) show distinct temporal evolution (Figure 4B), corresponding to carrier lifetimes of 460.3, 379.9, 294.6, and 338.0 ps, respectively. The time-resolved near-field images also exhibit spatially nonuniform dynamics of the nanostructure (Figure S6). Figure 4C presents the time-resolved s-SNOM map along the nanowire (dashed line in Figure 4A). The carrier lifetime extracted from the spatiotemporal mapping is strongly correlated to the topography, as evidenced by the qualitatively similar profiles (Figure 4D). The nonuniform carrier lifetime is mediated by the spatial heterogeneity in the silicon nanostructure.³⁸ The mapping of carrier dynamics at a spatial resolution of 35 nm is demonstrated. These results further indicate that our near-field ultrafast nanoscopy can probe the local doping, defects, and barriers at the nanoscale in semiconductor materials and devices.

In summary, we have demonstrated visible-near-infrared pump–probe near-field nanoscopy as a noninvasive tool to examine the carrier dynamics in silicon nanostructures. The combination of ultrafast optics and near-field imaging permits the investigation of carrier kinetics with both high temporal (picosecond) and spatial (sub-50 nm) resolutions. In addition to silicon, the use of a visible pump beam enables the investigation of photocarrier dynamics in a wide range of semiconductor materials that are commonly used in practical applications, including germanium and III–V materials. This capability is valuable for the characterization and optimization of functional optoelectronic devices. While this work analyzed the carrier dynamics in SiNWs as a demonstration, the proposed pump–probe near-field nanoscopy can serve as a versatile and general optical diagnostic platform to study nanomaterials, including two-dimensional materials^{39,40} and quantum dots.⁴¹ It also provides a promising tool to study other nonequilibrium thermodynamic phenomena in nanomaterial systems, including phase transitions,^{42–44} energy and charge transfer,⁴⁵ and phonon propagation.⁴⁶

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.nanolett.2c04790>.

Experimental Details section including sample preparation and optical setup; Supporting Notes 1–4, Supporting Table 1, and Supporting Figures 1–6 on additional data and analysis (PDF)

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Author Contributions

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Notes

The authors declare no competing financial interest.

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